

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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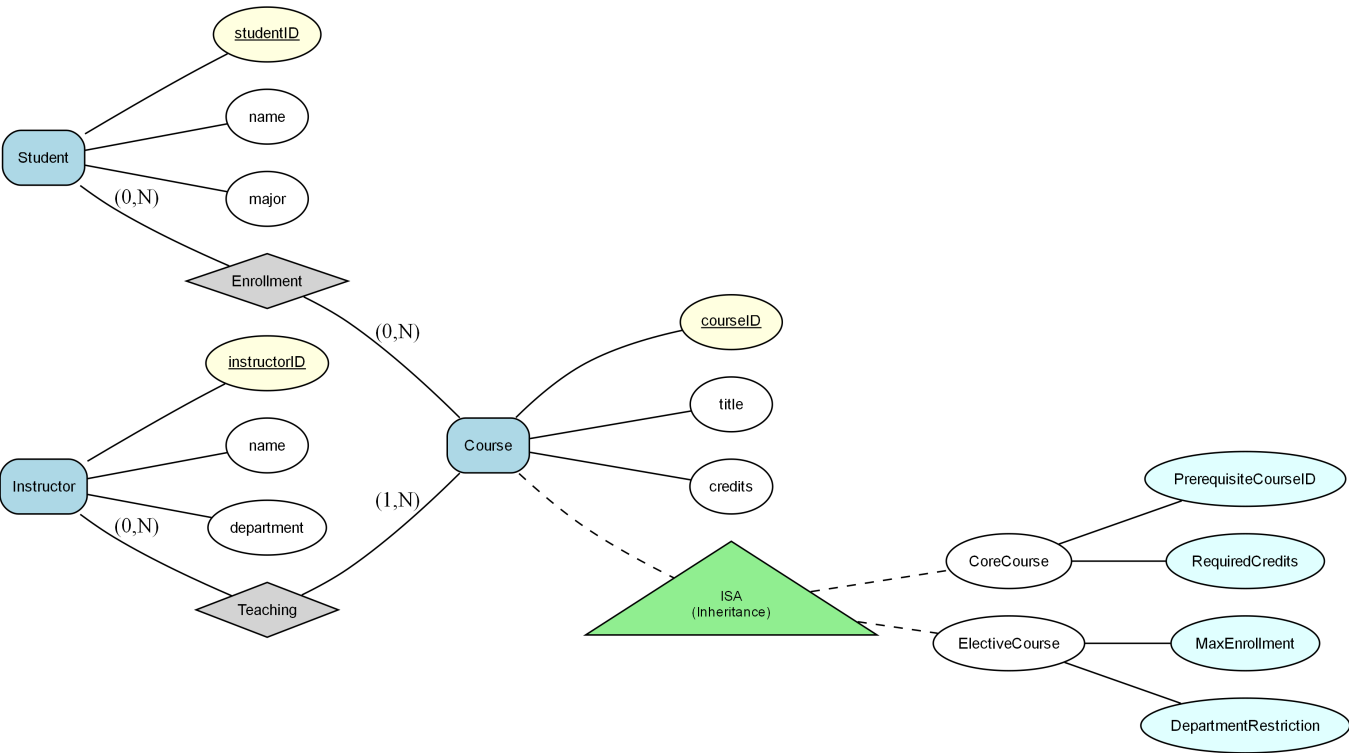
Total Marks: 100

Examination Paper

Question Q1 (20 marks)

A university maintains a database to manage its courses, students, and instructors. Each course has a unique course ID, a title, and a number of credits. Students have a student ID, name, and major. Instructors are defined by an instructor ID, name, and department. Each course can have multiple students enrolled, and each course is taught by one instructor. Additionally, students can take multiple courses. Model this scenario using an Enhanced Entity-Relationship (EER) diagram that incorporates courses, students, instructors, and their relationships.

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3)

Question Q2 (15 marks)

Consider a relation $R(\text{CustomerID}, \text{Name}, \text{Address}, \text{OrderID}, \text{Product}, \text{Quantity})$ with the following set of functional dependencies F over R : $F=\{\text{CustomerID} \rightarrow \text{Name}, \text{OrderID} \rightarrow \text{Product}, \text{Product} \rightarrow \text{Quantity}\}$. Analyze

the relational schema for normalization.

- a) Draw the functional dependency diagram. (1)
- b) Use the attribute closure and identify the primary key for the given relation. (1)
- c) If we insert a row <102, John Doe, 123 Main St, 201, Laptop, 3> into the given relation, what is the update anomaly that is violated? Give reasons. (2)
- d) Assume that the relation has a row <101, Jane Smith, 456 Elm St, 200, Phone, 1>. If we remove the order with OrderID 200, what is the possible update anomaly that can be violated? Give reasons. (2)
- e) Using the above functional dependencies, design a set of 3NF relations for the above given relation. Show clearly each stage in deriving the 3NF relations. Also identify the normal form at each of these stages. (3)
- f) Do the update anomalies of part (c) and (d) hold in the 3NF relations? Discuss briefly. (2)
- g) Are the 3NF relations derived in part (e) still in 3NF? Justify your answer. (2)
- h) Are the 3NF relations derived in part (e) in BCNF? If not bring the relations to BCNF. (2)

Question Q3 (25 marks)

A retail organization is establishing a new relational database to keep track of its inventory and customer orders. The database will consist of multiple tables including Products, Customers, and Orders. Each table will have specific attributes such as ProductID, ProductName, CustomerID, CustomerName, OrderID, and OrderDate. The organization seeks to ensure that all data adheres to certain integrity constraints and best practices in database design.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2)

b) Code Segment:

```
```sql
PreparedStatement pstmt = connection.prepareStatement(
 "SELECT * FROM Employees WHERE Department = ?");
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2)

- c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2)
- d) Briefly explain how authentication and authorization are achieved in SQL Server. (3)
- e) A retail organization has designated a user named Alice as the primary database administrator responsible for managing inventory data. Write a T-SQL statement to create a login to Alice with Windows authentication. (3)
- f) Provide Alice with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3)
- g) Assuming Bob's username is 'bob.m', write T-SQL statements to assign him the responsibility of managing the Orders table using a user-defined server role. (4)
- h) Assuming Charlie's login name is 'charlie.c', write T-SQL statements to grant him permission to select and insert records in the Products table. (3)
- i) Assuming David's login name is 'david.d', write T-SQL statements to allow him to perform updates on the Customer table. (3)

Question Q4 (40 marks)

A university department maintains a database to manage its courses and enrollments. The relation `Course(course_id: string, course_name: string, credits: integer, department: string)` contains details of all courses offered. The relation `Enrollment(student_id: string, course_id: string, semester: string)` records which students are enrolled in which courses during a particular semester. Based on this schema, answer the following questions.

- a) Find the names of the courses that have more than 3 credits. (5)
- b) Find the student IDs for those who are enrolled in the 'Database Systems' course. (5)
- c) Get the details of courses offered in the 'Computer Science' department during the 'Fall 2024' semester. (8)
- d) Retrieve the names of students who are enrolled in all courses offered in the 'Spring 2024' semester. (10)
- e) How many students are enrolled in the 'CS101' course? (4)
- f) Express the above queries (a) to (e) in tuple calculus. (8)